

Stoquify Enterprise Sales-to-Cash Implementation Roadmap

Roadmap date: 2026-08-17

Source: STOQUIFY_ENTERPRISE_SALES_TO_CASH_AUDIT_AND_MODERNIZATION_2026-08-16

Current program status: **BLOCKED — implementation and production proof required**

Planning horizon: 32–40 weeks with two dedicated delivery squads and shared control reviewers

Classification: implementation program; not a legal, accounting, PCI, security, accessibility, or production certification

1. Roadmap decision

Stoquify should modernize sales-to-cash through two coordinated product paths:

1. Harden the existing immediate POS path until cash, electronic tender, stock, accounting, receipt/fiscal evidence, offline replay, shift close, and reconciliation are trustworthy under concurrency and failure.
2. Build a separate delivery/on-account order-to-cash orchestrator for promise, reservation, fulfillment, goods issue, invoicing, AR, collection, returns, and corrections.

Both paths must reuse the same inventory, valuation, accounting, provider-event, reconciliation, receipt/fiscal, audit, close-invalidated, and outbox kernels. The program must not create a second immediate-sale finalizer.

The implementation sequence is:

```
flowchart LR
    P0["Phase 0<br/>Program and contract freeze"] --> P1["Phase 1<br/>Transaction and access foundations"]
    P1 --> P2A["Phase 2A<br/>Provider-authoritative payments"]
    P1 --> P2B["Phase 2B<br/>Immutable receipt/fiscal source"]
    P2A --> P3["Phase 3<br/>Immediate POS controlled release"]
    P2B --> P3
    P3 --> P4["Phase 4<br/>Business day and statement close"]
    P1 --> P5["Phase 5<br/>Delivery order and fulfillment"]
    P5 --> P6["Phase 6<br/>Invoice, AR, returns and corrections"]
    P4 --> P7["Phase 7<br/>Unified operations and hardening"]
    P6 --> P7
    P7 --> P8["Phase 8<br/>Country pilot and production proof"]
```

No phase is complete because code exists. It is complete only when its state contract, migrations, runtime behavior, negative authorization tests, accounting/inventory tie-outs, recovery behavior, operational evidence, and reviewer decisions pass the named exit gate.

2. Ambition and measurable outcomes

The completed program must provide:

- One authoritative owner for every material sales transition.

- Exactly-once observable outcomes for retries without pretending distributed processing is globally exactly once.
- Provider-owned electronic payment truth from initiation through settlement, reversal, dispute, and chargeback.
- Immutable receipt/fiscal-source evidence for every completed sale, independent of delivery success.
- Correct inventory quantity and COGS movement at the physical goods-issue event.
- Balanced, source-linked accounting for cash, provider clearing, AR, revenue, tax, COGS, inventory, fees, refunds, settlements, disputes, and corrections.
- Delivery/on-account order processing with reservation, partial fulfillment, invoicing, collection, return, and correction.
- Distinct and observable session close, tender declaration, business-day statement, provider reconciliation, and accounting close.
- Signed, ordered, replay-safe, conflict-aware, and operationally resolvable offline events.
- Consistent tenant/location isolation, RBAC, module entitlement, fresh authentication, redaction, audit, and separation of duties.
- Accessible cashier, manager, accountant, inventory, treasury, and support workflows.
- A release evidence pack tied to one commit, one environment, one provider configuration, and one qualified country pack.

2.1 Program-level acceptance metrics

Measure	Required outcome before controlled production
Completed-sale result coverage	100% of completed sales have an immutable result-registry record
Retry behavior	100% identical retries return the original result; all conflicting payloads reject and audit
Duplicate consequences	Zero duplicates in repeated PostgreSQL race and response-loss tests
Electronic payment truth	Zero sale completions from pending, unknown, timed-out, or operator-reference-only payment states
Receipt/fiscal source	100% of completed sales have an immutable source payload/hash before response
Stock/accounting tie-out	Every sale, fulfillment, correction, and return ties source quantity/cost to postings
Ledger balance	Every posting batch balances; posted history is corrected only through linked records
Provider reconciliation	100% of captured funds are settled, reversed, refunded, disputed, or in owned suspense
Close integrity	No final business-day statement with unresolved offline, drawer, provider, fiscal, or posting blockers
Isolation	Zero cross-tenant and unauthorized cross-location access in the full negative matrix
Sensitive data	Zero PAN/CVV/PIN/secrets or unnecessary customer PII in application logs and evidence
Accessibility	Critical cashier/manager tasks pass the approved keyboard, focus, screen-reader, touch, and responsive matrix
Recovery	Every queued, unknown, conflict, and dead-letter state has a tested owner and runbook

3. Non-negotiable implementation constraints

- No destructive database reset or destructive migration.

- No mutation or deletion of completed sales, payments, stock movements, receipts, fiscal documents, journals, reconciliation evidence, or audit records.
- Financial and stock corrections use linked compensating records.
- All reads and writes remain organization-scoped and location-scoped where the business fact is location-bound.
- All critical commands enforce explicit RBAC and module entitlement on the server.
- High-risk commands require fresh authentication and maker-checker approval where material.
- Receipt delivery failure never rolls back or conceals a completed sale.
- Provider uncertainty never becomes successful payment.
- Offline clients never allocate final fiscal numbers without country-pack authority.
- Logs and evidence exclude secrets, PAN, CVV, PIN, raw provider payloads, and unnecessary customer PII.
- Shared code contains no hard-coded national tax, receipt, numbering, or retention law.
- Country-pack conclusions require dated, versioned, qualified-human review.
- Existing dirty-worktree changes must be preserved and attributed.
- `commitPOSSale` remains the only immediate-sale finalizer.
- Analytics, loyalty, CRM, forecasting, and recommendation work remains deferred until transaction truth is complete.

4. Delivery model and capacity assumptions

The 32–40 week horizon assumes:

- Squad A — Transaction Trust and Payments: POS finalization, idempotency, provider state, receipts, offline, and close.
- Squad B — Order-to-Cash and Store Operations: order, reservation, fulfillment, invoice, AR, returns, business day, and operator experience.
- Shared reviewers — accounting controller, inventory controller, payments/treasury, security, country-pack expert, accessibility, operations/support, SRE/release, and product.
- A real PostgreSQL integration environment and at least one provider sandbox are available by Phase 2.
- One pilot country, currency, provider set, and receipt/fiscal configuration are chosen by Phase 0.
- Hardware profiles for scanner, printer, drawer, and terminal are available by Phase 3.

If only one delivery squad is available, plan 48–60 weeks. Adding developers without dedicated controller, security, provider, country-pack, accessibility, and SRE decisions will not shorten the critical path.

5. Program workstreams

Workstream	Scope	Accountable owner	Principal dependencies
WS-01 Transaction integrity	Client result registry, canonical hashes, concurrency, recovery	Principal backend/POS engineer	Architecture and migration policy
WS-02 Access and fraud controls	Permission vocabulary, entitlement, fresh auth, maker-checker, redaction	Security architect	Product risk thresholds
WS-03 Payments and treasury	Provider state, webhook/polling, clearing, settlement, fees, suspense, disputes	Payments lead/treasury	WS-01, provider sandbox

Workstream	Scope	Accountable owner	Principal dependencies
WS-04 Receipt and fiscal evidence	Immutable source, country adapter, numbering, delivery, correction documents	Compliance engineering lead	WS-01, country-pack reviewer
WS-05 Inventory and fulfillment	Reservation, availability, pick/ship, goods issue, valuation, returns	Inventory domain lead	Order contract and product policy
WS-06 Accounting and AR	Posting rules, invoices, AR, allocation, credit notes, close invalidation	Accounting engineering lead/controller	WS-03, WS-05
WS-07 Store close and reconciliation	Session, declaration, business day, statement, X/Z, sign-off, handoff	Store operations lead	WS-03, offline evidence
WS-08 Cashier and manager experience	Checkout, recovery, hardware, offline, accessibility, privacy, handover	Product/UI lead	Frozen domain contracts
WS-09 Reliability and release	Observability, SLOs, load, restore, rollout, rollback, evidence pack	SRE/release lead	All workstreams

6. Critical path and parallelization

Critical path:

1. Contract and control freeze.
2. Result idempotency and access normalization.
3. Provider-authoritative payment and immutable receipt source.
4. Immediate POS controlled release evidence.
5. Business-day statement and reconciliation.
6. Unified operational hardening.
7. Country pilot and production proof.

Delivery/on-account can begin after Phase 1 and run in parallel with store-close work:

```

flowchart TD
    A["G0: Contracts approved"] --> B["Result registry"]
    A --> C["Access normalization"]
    B --> D["Provider payment bridge"]
    B --> E["Immutable fiscal source"]
    C --> D
    C --> E
    D --> F["POS controlled release"]
    E --> F
    F --> G["Business-day statement"]
    B --> H["Order aggregate"]
    C --> H
    H --> I["Reservation and fulfillment"]
    I --> J["Invoice, AR and corrections"]
    G --> K["Unified operations"]
    J --> K
    K --> L["Country pilot"]
  
```

Parallel work is allowed only when teams own disjoint files or interfaces and consume frozen contracts. Two teams must not independently change sale finalization, payment-state normalization, posting rules, or fiscal-source contracts.

7. Phase plan

Phase 0 — Program control and contract freeze

Indicative duration: weeks 1–2

Entry: audit accepted as the planning baseline

Exit gate: G0 Architecture and Control Freeze

Objectives

- Convert audit decisions into versioned implementation contracts.
- Select the first pilot country, currency, locations, provider(s), and hardware profile.
- Establish a clean program baseline without discarding the current dirty worktree.
- Define evidence ownership and stop/go authority.

Work packages

ID	Deliverable
FND-01	Program charter, scope, non-goals, decision register, dependency register, and named owners
FND-02	State-transition catalog for sale, provider payment, receipt/fiscal source, session, statement, order, reservation, fulfillment, invoice, payment allocation, return, and correction
FND-03	Command contract template: actor, permission, entitlement, precondition, transaction, idempotency, event, accounting, inventory, audit, recovery, terminal state
FND-04	Additive migration and backfill policy; rollback through feature flags/shadow paths rather than data destruction
FND-05	Canonical money, currency minor-unit, tax snapshot, date/time/business-day, and location rules
FND-06	Evidence index and release-gate ownership
FND-07	Dirty-worktree attribution and branch/worktree coordination plan

Mandatory decisions

- Product: immediate tender types, on-account policy, delivery flow, partial fulfillment, return/refund scope.
- Accounting: posting roles, recognition timing, clearing, AR, fees, tax, corrections, close.
- Inventory: availability, reservation, goods issue, negative stock, valuation, return disposition.
- Payments: provider canonical-state mapping, unknown handling, settlement and dispute policy.

- Country pack: required receipt/fiscal fields, numbering, offline/provisional behavior, retention, corrections.
- Security: permission vocabulary, high-risk thresholds, maker-checker rules, evidence retention.

G0 exit criteria

- Every material transition has one proposed authoritative owner.
- State and command contracts are approved by the relevant controller.
- The first country/provider/hardware pilot scope is explicit.
- Unsupported multi-currency, tax, UOM, lot/serial, provider, and fiscal combinations fail closed.
- A clean named implementation baseline is available; unrelated user changes remain preserved.

Phase 1 — Transaction and access foundations

Indicative duration: weeks 3–6

Entry: G0 passed

Exit gate: G1 Transaction Trust

Stream 1A: tenant-scoped POS result registry

ID	Implementation slice
TRN-01	Add result-registry schema keyed by organization, terminal, and <code>clientCommitId</code>
TRN-02	Add canonical request hashing and schema version
TRN-03	Claim/result lifecycle with immutable redacted result envelope
TRN-04	Integrate into the existing <code>commitPOSSale</code> transaction
TRN-05	Return the original result for identical retries; reject and audit payload conflicts
TRN-06	Add real PostgreSQL race, response-loss, timeout, rollback, and replay tests
TRN-07	Bind sale, payment/AR, stock, journal, fiscal source, outbox, and result IDs

Stream 1B: access-boundary normalization

ID	Implementation slice
SEC-01	Freeze canonical POS/sales/payment/close permission vocabulary and compatibility aliases
SEC-02	Enforce module entitlement at route, page-data, read, command, offline, receipt, report, export, and API boundaries
SEC-03	Add organization/location negative test matrix
SEC-04	Replace over-privileged cashier seed behavior with least-privilege fixtures
SEC-05	Add fresh-auth and maker-checker policies for refunds, voids, overrides, payouts, variances, manual matches, goods-issue reversals, and credit notes

ID	Implementation slice
SEC-06	Add sensitive-data redaction and evidence-retention tests
SEC-07	Ratchet module-surface findings to prevent new unmapped or unenforced surfaces

G1 exit criteria

- Concurrent identical attempts create exactly one authoritative sale result and consequence set.
- Conflicting uses of the same client ID fail closed and audit.
- Failed transactions leave no completed registry result or partial financial/stock consequence.
- Navigation, reads, commands, offline actions, receipt actions, reports, and exports enforce approved authority.
- Cross-tenant and unauthorized cross-location tests pass.
- No destructive migration and no mutation of completed history.

Phase 2A — Provider-authoritative electronic payments

Indicative duration: weeks 7–12

Entry: G1 passed and provider sandbox available

Exit gate: G2A Payment Truth

Work packages

ID	Implementation slice
PAY-01	Versioned mapping from each provider state to canonical initiated, pending, authorized, captured, unknown, declined, cancelled, expired, settled, reversed, refunded, disputed, and chargeback states
PAY-02	Payment-intent command with provider and internal idempotency
PAY-03	Signed webhook and polling ingestion with timestamp, replay, signature, payload-hash, and monotonic-transition controls
PAY-04	POS checkout bridge that completes only after provider-authoritative capture
PAY-05	Captured-without-sale and late-capture exception/suspense workflows
PAY-06	Provider clearing, settlement, fee, reversal, refund, dispute, and chargeback postings
PAY-07	Statement import and match linkage to the same POS payment transaction and ledger source
PAY-08	Provider outage, duplicate callback, mismatch, key rotation, delayed statement, and retry runbooks

G2A exit criteria

- Pending, unknown, timeout, and manual reference never complete a sale.
- Duplicate webhooks and polls do not duplicate payment, sale, posting, or receipt consequences.
- Amount/currency mismatch and impossible state changes quarantine.
- Captured-without-sale, sale-without-settlement, settlement shortfall, fee, reversal, dispute, and chargeback remain visible until terminal resolution.
- Provider events, internal transaction, sale, clearing posting, statement line, and reconciliation certificate share traceable source links.
- Sandbox evidence covers success and all named failure paths.

Phase 2B — Immutable receipt and fiscal-source truth

Indicative duration: weeks 7–11, parallel with Phase 2A

Entry: G1 passed and country-pack review initiated

Exit gate: G2B Document Truth

Work packages

ID	Implementation slice
FIS-01	Canonical immutable receipt/fiscal-source payload and schema version
FIS-02	Source materialization and hash inside the existing sale transaction
FIS-03	Resolve unused synchronous fiscal helper; retain one fiscalization owner
FIS-04	Country-pack-governed legal number allocation and authority submission
FIS-05	Worker lease, retry, backoff, dead-letter, replay, and operator recovery
FIS-06	Durable print, email, SMS, WhatsApp, and public-token delivery states for supported channels
FIS-07	Token signing, expiry, scope, revocation, rotation, and release-secret enforcement
FIS-08	Linked credit/void/correction fiscal documents without rewriting originals

G2B exit criteria

- Every completed sale has immutable source evidence before its result returns.
- Later catalog, price, tax, customer, organization, or location edits do not change the receipt source.
- Legal numbering remains unavailable offline and fails closed without country-pack authority.
- Delivery failure is visible but never changes completed financial truth.
- Worker retry and dead-letter drills preserve one document/number per authorized scope.
- Production-secret configuration passes in release mode.
- Qualified country-pack review is dated and versioned; no shared code hard-codes national law.

Phase 3 — Immediate POS controlled-release completion

Indicative duration: weeks 13–16

Entry: G2A and G2B passed

Exit gate: G3 Immediate POS Release Candidate

Work packages

ID	Implementation slice
POS-01	Remove or disable UI tender methods the service cannot truthfully support
POS-02	Apply country/currency minor-unit and rounding contract to display, input, receipt, posting, and tests
POS-03	Connect offline checkout to the signed offline event and result-recovery flow
POS-04	Integrate supported scanner, receipt printer, cash drawer, and terminal health contracts
POS-05	Add inactivity lock, cashier handover, controlled session recovery, and safe refresh/resume
POS-06	Complete refund/void/receipt recovery and manager-approval interfaces
POS-07	Complete mobile cart, keyboard, focus, screen-reader announcements, touch targets, localization, privacy masking, and error recovery
POS-08	Add end-to-end telemetry with redacted correlation IDs and operator-safe messages

G3 exit criteria

- Cash and supported electronic tenders pass browser, device, concurrency, failure, and recovery scenarios.
- Unsupported tender/currency/tax/hardware capabilities are not displayed as available.
- Sale completion can always recover its immutable original result and receipt source.
- Offline conflict cannot create silent duplicate finalization or final fiscal numbering.
- Critical cashier and manager flows pass accessibility and responsive review.
- No secrets, PAN/CVV/PIN, or unnecessary PII appear in logs, support views, receipts, or screenshots.

Phase 4 — Governed business day, statements, and close

Indicative duration: weeks 17–22

Entry: G3 passed; provider reconciliation and offline evidence available

Exit gate: G4 Store Financial Close

Work packages

ID	Implementation slice
CLS-01	Location/time-zone-aware business-day aggregate supporting cross-midnight trading
CLS-02	Session inclusion and immutable source-manifest hash
CLS-03	Blind cash/tender declaration with denomination evidence
CLS-04	Variance threshold, fresh-auth approval, and no-self-approval rules
CLS-05	X report for interim review and immutable final Z report
CLS-06	Retail statement calculation from included completed transactions
CLS-07	Statement posting status and source-linked journal evidence
CLS-08	Provider settlement/reconciliation and suspense status
CLS-09	Fiscal, offline, stock, receipt, and posting blocker aggregation
CLS-10	Accounting-close handoff and invalidation after late postings/corrections

G4 exit criteria

- Session close, declaration, business day, statement, provider reconciliation, and accounting close have separate states and owners.
- A final Z/statement cannot omit accepted offline events or unresolved replay conflicts.
- Declared, system, provider, ledger, and bank totals are explainably tied or in owned suspense.
- Manager sign-off is supported, attributable, fresh-authenticated, and separation-of-duties compliant.
- Corrections after statement or close create linked evidence and invalidate affected readiness.
- X/Z naming and legal meaning are country-pack reviewed rather than assumed.

Phase 5 — Delivery/on-account order and fulfillment

Indicative duration: weeks 17–24, parallel with Phase 4

Entry: G1 passed; order/inventory/accounting contracts approved

Exit gate: G5 Physical Fulfillment Truth

Work packages

ID	Implementation slice
OTC-01	Add explicit order header/line version, terms, delivery location, price/tax snapshot, credit policy, and command idempotency
OTC-02	Confirm order as a non-posting promise
INV-01	Add reservation aggregate with active, partial, consumed, released, and expired states
INV-02	Calculate sellable availability from on-hand, reserved, policy, and location
FUL-01	Add fulfillment aggregate and partial release/pick/pack/ship/handover states
FUL-02	Post goods issue through the existing inventory stock-event/valuation kernel

ID	Implementation slice
FUL-03	Consume reservation exactly once and preserve backorder quantities
FUL-04	Add cancellation before goods issue and linked goods-issue reversal afterward
FUL-05	Add delivery evidence, proof of handover, and operator recovery
CAP-01	Fail closed for unimplemented UOM, lot, serial, expiry, negative-stock, or multi-location combinations

G5 exit criteria

- Order confirmation posts no revenue, AR, COGS, or inventory.
- Reservation changes reserved/available but not on-hand or COGS.
- Physical goods issue changes quantity and posts COGS/inventory exactly once.
- Partial fulfillment, backorder, cancellation, and reversal preserve original line history.
- Every stock event links to order line, fulfillment line, location, valuation evidence, actor, and command key.
- Concurrent reservation and goods-issue tests prove no oversell beyond approved policy.

Phase 6 — Invoice, AR, collections, returns, and corrections

Indicative duration: weeks 25–30

Entry: G5 passed; controller billing policy approved

Exit gate: G6 Financial Order-to-Cash

Work packages

ID	Implementation slice
AR-01	Build invoice eligibility from delivered quantities or explicit approved advance policy
AR-02	Post immutable invoice with AR, revenue, tax, posting rule, fiscal source, and source links
AR-03	Support partial invoices, due dates, allocations, partial payments, and unapplied cash
AR-04	Connect provider collections to the Phase 2 payment/reconciliation kernel
COR-01	Add return authorization and received/inspected/disposition states
COR-02	Add restock, quarantine, damaged, and write-off inventory consequences
COR-03	Add source-linked credit notes, refunds, AR adjustments, and goods-issue reversals
COR-04	Add disputes, chargebacks, and recovery cases without rewriting original sale/invoice/payment
AR-05	Add customer statement, aging, collection evidence, and accounting-close invalidation

G6 exit criteria

- Only eligible quantities are invoiced and no line is billed twice.
- Invoice posting balances AR/revenue/tax and binds immutable source/fiscal evidence.
- Customer payments allocate explicitly; unapplied or mismatched funds remain visible.
- Returns require disposition before sellable stock increases.

- Credit note, refund, stock correction, provider correction, and journal reversal all link to the original fact.
- Partial fulfillment/invoice/payment/return scenarios tie order, stock, AR, provider, and ledger quantities/amounts.

Phase 7 — Unified operations, maintainability, and hardening

Indicative duration: weeks 31–34

Entry: G4 and G6 passed

Exit gate: G7 Operational Readiness Candidate

Work packages

ID	Implementation slice
OPS-01	SLOs for checkout, provider callback, fiscal worker, delivery, offline replay, statement posting, reconciliation, fulfillment, invoicing, and receipt delivery
OPS-02	Redacted dashboards, alert routing, dead-letter queues, suspense aging, and ownership
OPS-03	Runbooks for every non-terminal/unknown/conflict state
OPS-04	Load, soak, concurrency, network-loss, provider-outage, queue-backlog, and recovery tests
OPS-05	Backup/restore, point-in-time recovery, failover, and evidence restoration drill
OPS-06	Decompose oversized POS component/service behind frozen contracts without changing business ownership
OPS-07	Accessibility regression, browser/device matrix, visual regression, performance budgets, and localization
OPS-08	Security scan, threat-model review, abuse tests, secret rotation, log/evidence review, and retention controls
OPS-09	Support timelines linking sale, result, receipt, payment, stock, journal, fulfillment, statement, and correction evidence

G7 exit criteria

- Every alert has an owner, severity, response time, runbook, and terminal resolution.
- Backup/restore and failure drills preserve immutable/source-linked evidence.
- Performance and queue backlogs remain within approved budgets under expected peak and degraded-provider load.
- Decomposition does not introduce a second finalizer or divergent posting/stock/fiscal logic.
- No unresolved critical security, isolation, accessibility, accounting, inventory, payment, or recovery finding.

Phase 8 — Country pilot and production proof

Indicative duration: weeks 35–40

Entry: G7 passed and qualified country/provider approvals available

Exit gate: G8 Production Expansion Decision

Rollout stages

Stage	Cohort	Entry evidence	Automatic rollback/stop condition
R0	Local and CI	All focused unit/integration/gate suites	Any invariant failure
R1	Staging with provider/fiscal sandbox	Full end-to-end and recovery matrix	Duplicate consequence, unexplained posting/stock mismatch
R2	Internal staff/demo organization	Controlled hardware and store day	False payment success, cross-tenant access, unrecoverable close
R3	One design partner, one country/provider/location set	Human approvals and daily evidence review	Any critical financial, fiscal, security, stock, or privacy issue
R4	Up to 5% eligible cohort	Stable SLOs and no unresolved material suspense	Threshold breach or unexplained cohort divergence
R5	Up to 25% eligible cohort	Two stable business-day/settlement cycles minimum	Close/reconciliation drift, support overload, material accessibility regression
R6	General availability by eligible country/package	Formal G8 decision and rollback readiness	Ongoing error-budget or control breach

Production evidence pack

- Named commit, build, environment, configuration, migration hashes, and deployed country-pack/provider versions.
- Additive migration status, backfill/reconciliation certificate, and rollback/disable plan.
- PostgreSQL concurrency and failure-injection evidence.
- Provider sandbox/live pilot webhook, poll, capture, settlement, fee, refund, reversal, dispute, and statement evidence.
- Receipt/fiscal worker, legal number, delivery, token secret, retry, dead-letter, and correction evidence.
- Inventory quantity/valuation/COGS tie-outs.
- Cash, clearing, AR, revenue, tax, COGS, inventory, fee, refund, settlement, and correction ledger tie-outs.
- Session, tender declaration, business-day, X/Z, statement, provider reconciliation, and accounting-close evidence.
- Tenant/location isolation, RBAC, entitlement, fresh auth, maker-checker, redaction, and audit evidence.
- Browser, accessibility, supported hardware, offline, performance, restore, failover, monitoring, and incident evidence.
- Dated reviewer decisions and explicit remaining limitations.

G8 exit decision

G8 can authorize only the tested country, currency, provider, hardware, tax, receipt/fiscal, inventory, and product combination. It must not imply support for an untested combination or certify PCI, legal, accounting, security, or accessibility status without the relevant qualified assessment.

8. Stage-gate governance

Gate	Decision	Required approvers	May not self-certify
G0	Architecture/control freeze	Product, architecture, accounting, inventory, payments, security, country-pack	Implementing engineer
G1	Transaction trust	POS/backend, DBA, security, accounting/inventory reviewers	Result-registry author alone
G2A	Payment truth	Payments, treasury, accounting, security	Provider adapter author alone
G2B	Document truth	Compliance engineering, country-pack expert, accounting, security	Shared-code author alone
G3	Immediate POS release candidate	Product, operations, accessibility, security, SRE	POS squad alone
G4	Store financial close	Store operations, treasury, accounting controller	Cashier/declarer
G5	Fulfillment truth	Inventory controller, product, accounting	Fulfillment squad alone
G6	Financial O2C	Accounting controller, treasury, inventory, product	O2C squad alone
G7	Operational candidate	SRE, security, accessibility, support, all controllers	Release engineer alone
G8	Production expansion	Executive product owner plus all material domain approvers	Any single function

At each gate, the decision must be **PASS**, **PASS WITH EXPLICIT LIMITED SCOPE**, or **BLOCKED**. “Code complete,” “tests mostly pass,” and “works locally” are not gate outcomes.

9. Additive migration and legacy-data strategy

9.1 Migration rules

- Add new tables/columns/indexes without resetting or rewriting transaction history.
- Use expand/backfill/verify/enforce/contract sequencing.
- Introduce uniqueness only after duplicates and nulls are measured and resolved through evidence-backed records.
- Backfills must be resumable, idempotent, organization-scoped, chunked, observable, and hash-certified.
- New code must tolerate old and new rows during rollout.
- Rollback disables new writers/read paths; it does not delete new evidence or restore old mutable behavior.

9.2 Legacy truth classification

Legacy fact	Migration treatment
Completed sale without result registry	Create a historical result-reference record from existing immutable IDs/hashes; do not pretend it proves original client retry
Electronic payment with only operator reference	Classify as legacy/unverified or reconciled-by-evidence; never backfill as provider-captured without provider proof
Sale without FiscalDocument	Materialize from the best preserved source only if provenance is sufficient; otherwise mark evidence gap
Mutable receipt presentation	Preserve original available evidence and mark confidence; do not reconstruct legal facts silently
Existing journals/stock events	Link through additive source references; never rewrite or delete
Sales order lifecycle labels without transitions	Do not synthesize fulfillment history without evidence
Existing close/reconciliation reports	Preserve and version; do not retroactively certify unsupported sign-off

10. Test and evidence strategy

10.1 Test pyramid by consequence

Layer	Required coverage
Contract/unit	State transitions, money/tax rounding, permission decisions, provider mappings, hashes, redaction
Service integration	Real PostgreSQL transactions, unique constraints, CAS, rollback, source links, balanced postings
Concurrency/chaos	Duplicate clicks, response loss, simultaneous terminals, webhook/poll races, queue retries, worker lease expiry
Cross-domain scenario	Sale/payment/stock/ledger/receipt; order/fulfillment/invoice/AR; statement/reconciliation/close
Browser/accessibility	Cashier, manager, support, keyboard, focus, screen reader, touch, responsive, recovery
Provider/hardware	Sandbox/live pilot, scanner, printer, drawer, network loss, delayed callback, offline replay
Operations	Load/soak, backup/restore, failover, dead-letter, suspense, incident, rollback

10.2 Mandatory negative scenarios

- Cross-tenant and unauthorized cross-location ID guesses.
- Missing/expired module entitlement.
- Stale authentication and self-approval.
- Same client ID with different payload.
- Provider pending/unknown/timeout/late capture/duplicate/replayed/forged callback.
- Amount/currency/provider-account mismatch.
- Sale commit failure after provider capture.
- Fiscal worker unavailable or authority rejects.
- Receipt delivery provider unavailable.
- Offline device revoked, signature invalid, sequence gap, expired policy, conflicting payload.

- Concurrent stock reservation/issue and negative availability.
- Partial delivery, partial invoice, partial payment, partial return, and duplicate correction.
- Close attempted with open session, open drawer, offline conflict, unresolved provider suspense, fiscal dead letter, or unposted journal.
- Recovery after browser refresh, process crash, network loss, worker restart, and database retry.

10.3 Evidence quality

Every gate artifact must distinguish:

- Static source evidence.
- Unit/mock evidence.
- Real database/runtime evidence.
- Provider/hardware/browser evidence.
- Staging proof.
- Production pilot proof.
- Qualified-human review.

Evidence must include commit/configuration/environment identifiers, timestamps, commands, outputs, source hashes, redaction confirmation, unresolved limitations, and reviewer identity.

11. Requirement-to-phase traceability

Audit blocker	Owning phase(s)
1. No client-result replay	Phase 1
2. No PostgreSQL concurrency proof	Phases 1, 7, 8
3. Manual reference becomes paid	Phase 2A
4. Incomplete provider canonical state	Phase 2A
5. No provider-to-statement-to-ledger proof	Phases 2A, 4, 8
6. No immutable source in every completed sale	Phase 2B
7. Fiscal helper ownership ambiguity	Phase 2B
8. No production fiscal worker/authority proof	Phases 2B, 8
9. Production receipt token secret absent	Phases 2B, 8
10. Incomplete durable receipt channels	Phases 2B, 3
11. No qualified country-pack approval	Phases 0, 2B, 8
12. No delivery order-to-cash workflow	Phases 5, 6
13. No partial fulfillment/invoice/refund contract	Phases 5, 6
14. No UOM/lot/serial/expiry POS proof	Phases 0, 5; unsupported until proven
15. No business-day/X/Z/statement aggregate	Phase 4
16. Reconciliation/sign-off unsupported	Phase 4

Audit blocker	Owning phase(s)
17. Inconsistent permissions/entitlements	Phase 1
18. Broad module enforcement candidates	Phases 1, 7
19. Cashier least privilege/handover/SoD incomplete	Phases 1, 3, 4
20. Store-credit UI/service contradiction	Phase 3
21. XAF minor-unit policy unproven	Phases 0, 3, 8
22. Offline checkout not connected to UI	Phase 3
23. Hardware readiness synthetic	Phases 3, 8
24. Mobile/accessibility/privacy/recovery gaps	Phases 3, 7, 8
25. No production isolation/load/restore/observability proof	Phases 7, 8
26. No formal certifications	External qualified assessments after G7; never self-certified
27. Dirty worktree not tied to release commit	Phases 0, 8

All 27 audit blockers have an owning phase. A phase cannot close while its blocker remains unowned, silently deferred, or relabeled as complete without evidence.

12. Program risk register

Risk	Early warning	Mitigation	Owner
Provider semantics differ from assumed canonical states	Manual exceptions or state regressions increase	Versioned adapter, raw redacted evidence, quarantine, provider sandbox contract tests	Payments lead
Country-pack decision arrives late	Shared code starts accumulating jurisdiction logic	Choose pilot country at Phase 0; block fiscal claims; use fail-closed adapter	Product/compliance
Existing dirty changes collide with core work	Unattributed diffs or failing baselines	Named baseline, file ownership, narrow worktrees, frequent status attribution	Program lead
POS service remains a change hotspot	Merge conflicts and regression frequency	Freeze contracts, extract behind tests only after G3	Architecture lead
Backfill invents historical truth	Synthetic captured/fiscal/fulfillment states	Confidence classification, evidence gaps, no unsupported reconstruction	Data/accounting
O2C duplicates POS kernels	New posting/stock/receipt code appears in orchestrator	Architecture gate and dependency rules	Architecture/controller
Scope expands to full ERP parity	Phase deliverables grow without closing blockers	Non-goals and gate-owned change control	Product
Hardware/provider access blocks proof	Tests remain mocked late in program	Secure devices/sandboxes in Phase 0	Operations/payments
Maker-checker blocks small-business usability	Excessive approvals and abandonment	Materiality thresholds, role-aware policy, measured pilot	Product/security/controller
Evidence contains sensitive data	Raw callbacks or customer data appear in logs	Redaction schema, test fixtures, secret/DLP scanning	Security

Risk	Early warning	Mitigation	Owner
Team optimizes for passing static gates	Runtime failures remain undiscovered	Evidence taxonomy and real PostgreSQL/provider/browser gates	Release/SRE

13. Operating metrics after launch

Transaction integrity

- Sale attempts, registry claims, original-result replays, payload conflicts, and abandoned claims.
- Duplicate-consequence detector count: target zero.
- Commit latency and rollback/error rate by tender/location/terminal.

Payments and treasury

- Pending/unknown age, late captures, captured-without-sale cases, provider callback latency.
- Settlement match rate, fee variance, suspense count/value/age, disputes, and chargebacks.
- Every amount is measured by organization, provider account, currency, and business date.

Receipt/fiscal

- Completed-sale source coverage, fiscal queue age, authority rejection, dead letters.
- Delivery success/retry/failure by channel without exposing destination PII.
- Sequence gaps/duplicates and country-pack version distribution.

Inventory and O2C

- Reservation aging, oversell attempts, partial fulfillment, backorder age.
- Goods-issue-to-invoice lag, stock/COGS tie-out, return disposition age, write-off approvals.
- Order-to-cash cycle time and AR aging.

Store close

- Sessions closed on time, variance value/frequency, pending approvals.
- Business days blocked, blocker age, statement-posting lag, reconciliation completion.
- Late corrections and close invalidations.

Experience and reliability

- Checkout completion/recovery, offline conflicts, hardware failure, receipt recovery.
- Accessibility task success and critical regression count.
- SLO/error-budget, worker backlog, restore/failover results, support case resolution.

Metrics must support action and control; they must not become a second transaction truth.

14. Future Codex execution sequence

Each execution should be a bounded slice with its own ownership, baseline, tests, evidence, and handoff.
Recommended sequence:

1. Freeze sales-to-cash state, command, permission, money, and country-capability contracts.
2. Implement tenant-scoped POS client-result registry and real PostgreSQL concurrency proof.
3. Normalize POS RBAC/module entitlement, least-privilege roles, fresh auth, and maker-checker.
4. Implement canonical provider state and signed event/poll ingestion.
5. Connect POS electronic tender to provider capture, clearing, suspense, and result replay.
6. Implement immutable receipt/fiscal source in the sale transaction and durable delivery.
7. Complete immediate POS offline/hardware/recovery/accessibility experience.
8. Implement business day, blind declaration, X/Z, statement, reconciliation, and sign-off.
9. Implement delivery order confirmation and reservation.
10. Implement partial fulfillment, physical goods issue, reversal, and delivery evidence.
11. Implement invoice, AR, allocations, returns, credit notes, refunds, and disputes.
12. Implement unified observability, support timelines, SLOs, chaos, restore, and release evidence.
13. Run the limited country/provider/hardware pilot and close every blocker before expansion.

Every run must:

- Re-read this roadmap and the audit.
- Capture `git status --short` before work.
- Declare exact files and transitions owned.
- Avoid destructive migrations and preserve existing evidence.
- Add only the smallest additive schema required for the slice.
- Reuse existing kernels and reject second-owner designs.
- Run focused tests plus the relevant global gates.
- Produce a dated implementation/evidence report with local versus production proof.
- Leave unresolved decisions as explicit blockers rather than assumptions.

15. Definition of complete

The sales-to-cash ambition is fulfilled only when:

1. All G0–G8 decisions are passed for an explicit supported scope.
2. All 27 audit blockers are closed, deliberately unsupported with fail-closed behavior, or assigned to an externally qualified assessment.
3. Immediate POS and delivery/on-account workflows share kernels without sharing an overloaded lifecycle.
4. Every terminal financial, inventory, receipt/fiscal, payment, settlement, statement, and correction fact is immutable, source-linked, idempotent, scoped, authorized, auditable, and operationally recoverable.
5. The production evidence pack is bound to the deployed commit and environment.
6. Remaining limitations are visible to operators and customers and are not marketed as supported capability.

Until then, the appropriate status remains **implementation in progress — not production certified**.